



National University
of Computer & Emerging Sciences



Recommender System

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WELCOME TO THE COURSE

Course Description (Interesting Ingredient)

Teaching Material:

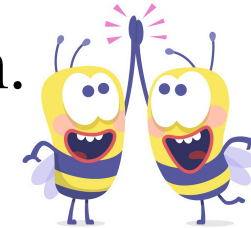
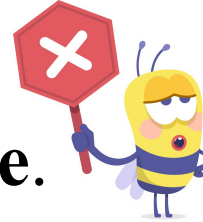
Textbook:

Francesco Ricci · Lior Rokach · Bracha Shapira ·
Paul B. Kantor
Editors

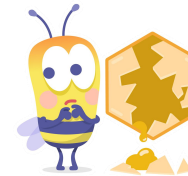
Recommender Systems Handbook

Class Room Policies – Pay attention

- Don't come into the class if you are **late**.
- Don't sit in my class for time passing or you have less interest.
- Attend the class with full motivation and passion.



Remember you are in the class to learn something new.



GOOGLE CLASSROOM



guwtmprz

<https://classroom.google.com/c/Nzk4ODM0NTI3ODQ3?cjc=guwtmprz>

ClassRoom Policies



Activities such as Lecture recording and taking Photos of the board are not allowed



Course Policies – Pay attention

- Assignments must be submitted with in *due dates*.
- Late submission will be subjected to the penalty which is as follow:

After **2** days of deadline **30 %** of marks deductions
After **4** days of deadline **40 %** of marks deductions
After **5** days of deadline **100 %** of marks deductions

- Student contact hours in my office are

Thursday 10:00 AM - 1:00 PM

Communication rules and Contact hours

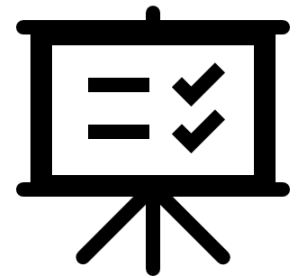
- For your queries related to the course, send an email to the following email address with clearly mention your **Class and Student ID** in the SUBJECT of an email.

mfarrukh.shahid@nu.edu.pk

Or you can visit the office (in contacting hours) **AI and Data Science Department**

Or discuss in the class (this is highly subjected to the time availability in a class)

*The **Course** objectives are follows,*



Course Objectives



- Have knowledge of the importance and benefits of RS
- Have knowledge and understanding of RS problems and their solutions
- Understand RS algorithms,
- Have knowledge of the commonly used tools and techniques of RS
- Understand how data must be pre-processed before application of RS algorithms
- Understand how to analyse and use the results obtained from the algorithms.

Course Objectives



- How **RS** has been changing the world ?
- From theory to the practical journey – Neuron -> **UAVs**, **Robots** etc.
- Insights of Model-driven and Data-Driven approaches.
- How to introduce intelligence into the devices of **IoT** network, **6G** mobile networks ?
- Give you full insight to select your MS Thesis.
- Developing International Collaboration
- Roadmap for Higher Education Abroad

Course Content

Recommendation System

Recommendation systems aim to predict users' preferences for a certain item and provide personalized services [1]

RSs analyze data to discern user preferences for items and assist users in efficiently sifting through information [2], directing them towards the content that is most relevant and valuable for their interests.

Information overload is a critical issue facing numerous e-Business, social networks (e.g., Youtube and Facebook), cloud-based, and mobile-based platforms [3]. Hence, the users' experience when searching for interesting items becomes challenging and time-consuming.

Recommender System (RS) is a common solution in such applications that offer users large amounts of products and services [3].

An RS provides users with a list of potentially interesting items; thus, it facilitates the shopping process and incurs benefits to businesses.

[1] B. Yi, X. Shen, H. Liu, Z. Zhang, W. Zhang, S. Liu, and N. Xiong, "Deep matrix factorization with implicit feedback embedding for recommendation system," IEEE Transactions on Industrial Informatics, vol. 15, no. 8, pp. 4591–4601, 2019

[2] Leiva, M.; Budán, M.C.D.; Simari, G.I. Guidelines for the analysis and design of argumentation-based recommendation systems. IEEE Intell. Syst. 2020, 35, 28–37.

[3] Bushra Alhijawi, Mutaz Abo Alrub, Mustafa Al-Fayoumi, Generalized Ethereum Blockchain-based recommender system framework, Information Systems, Volume 111, 2023, 102113, ISSN 0306-4379,

What do you think about RS



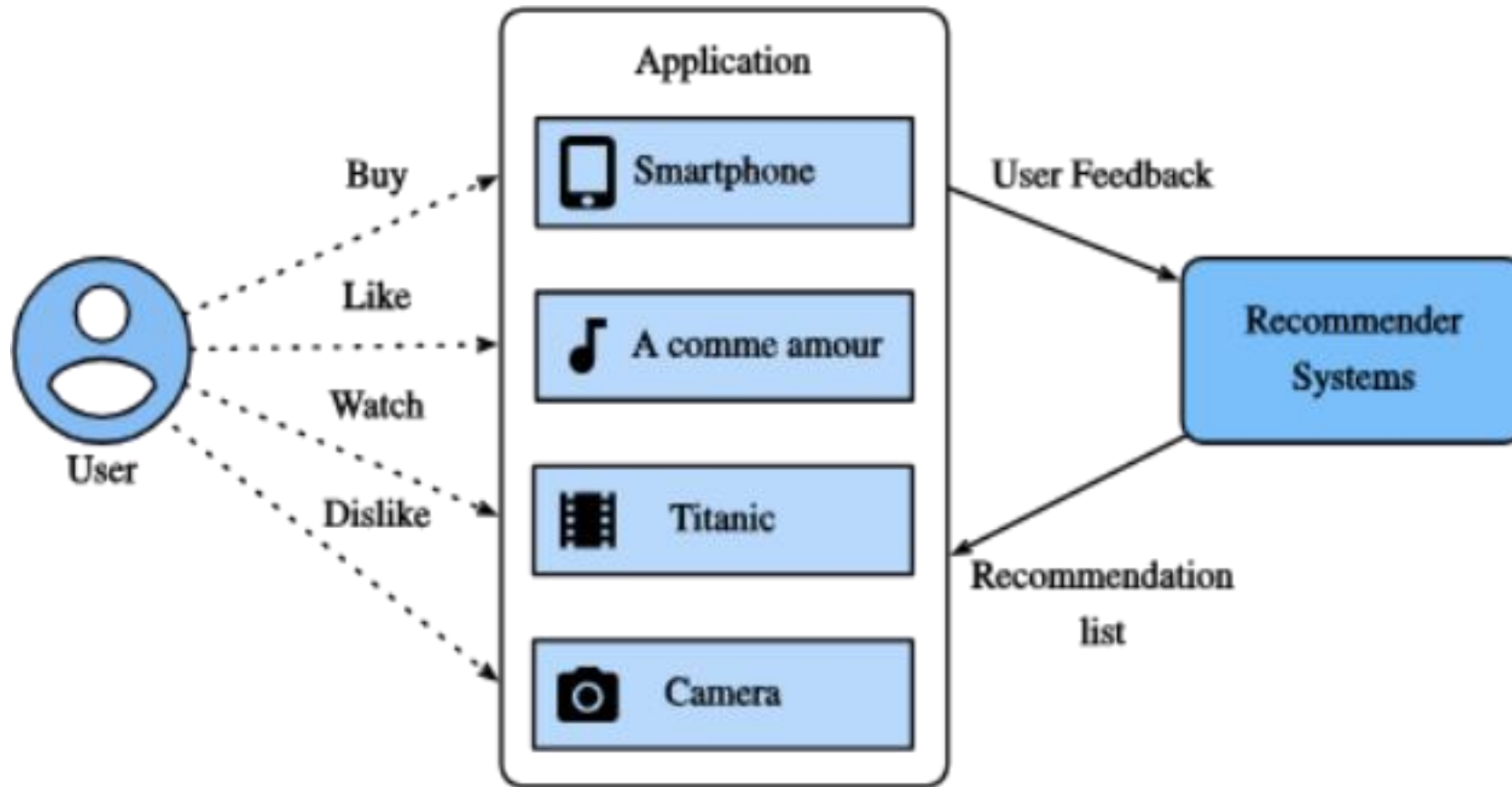
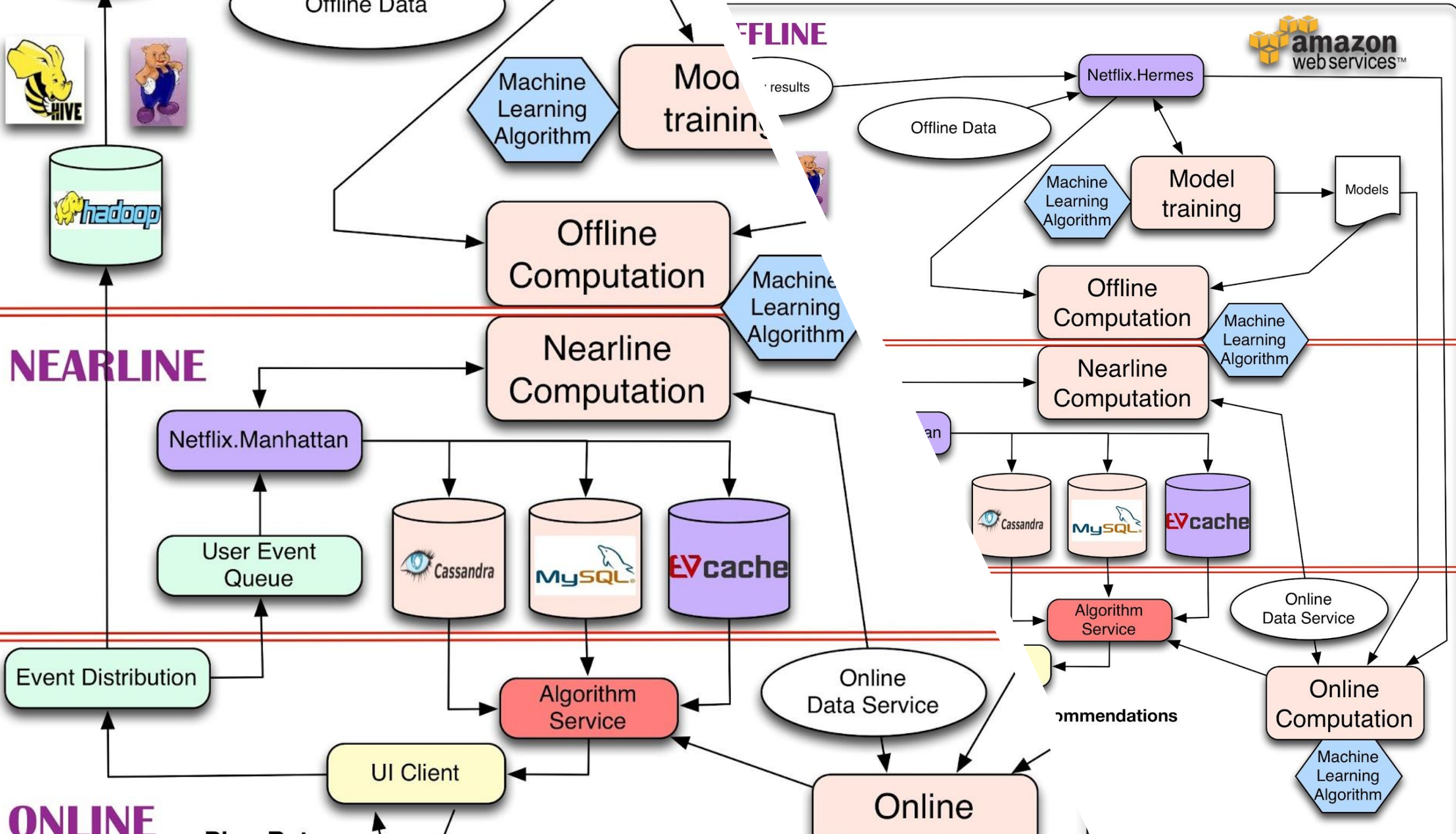
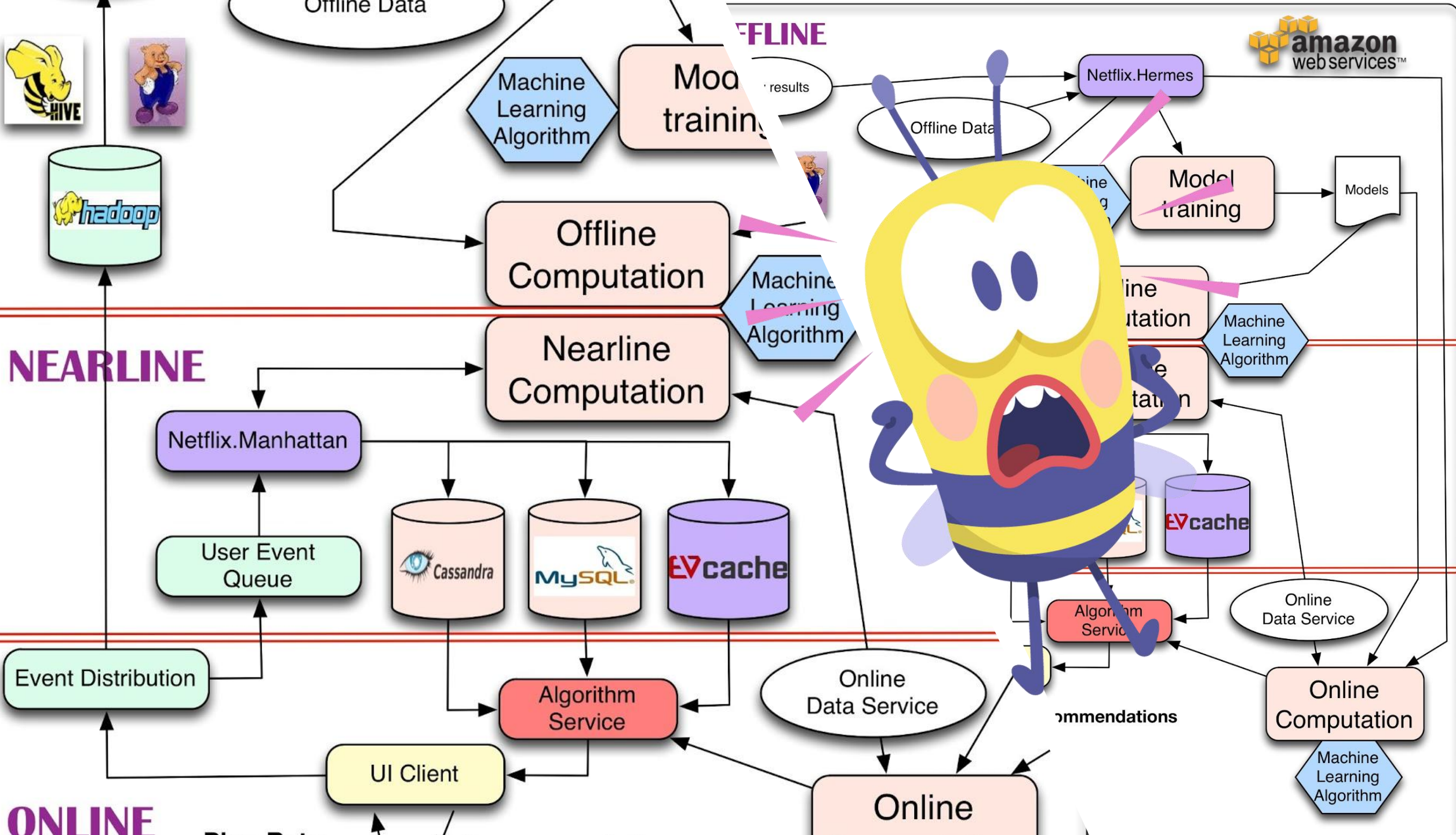


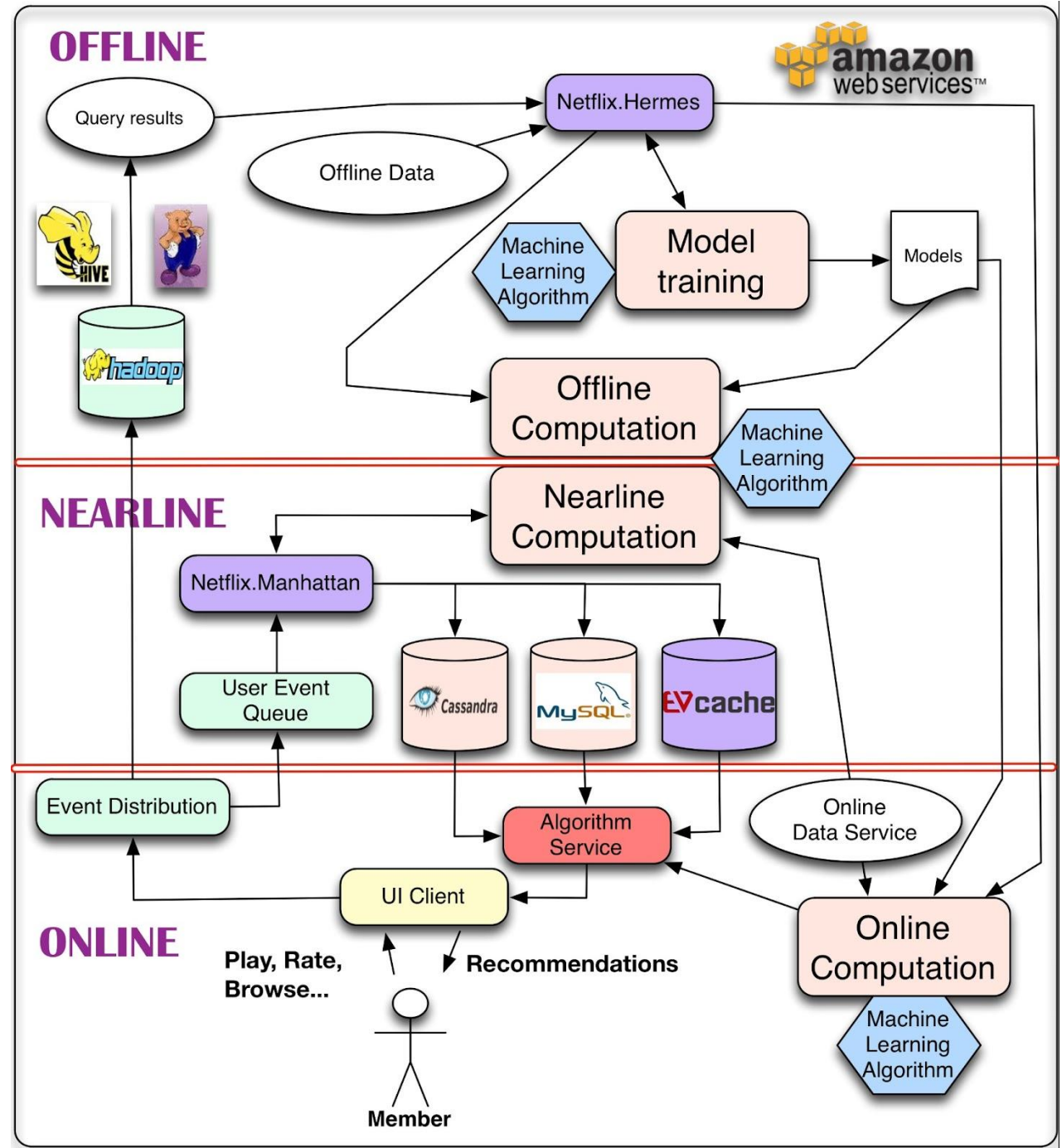
Fig. Illustration of the Recommendation Process [1]

What is RS in actual ?





The Netflix three tier architecture

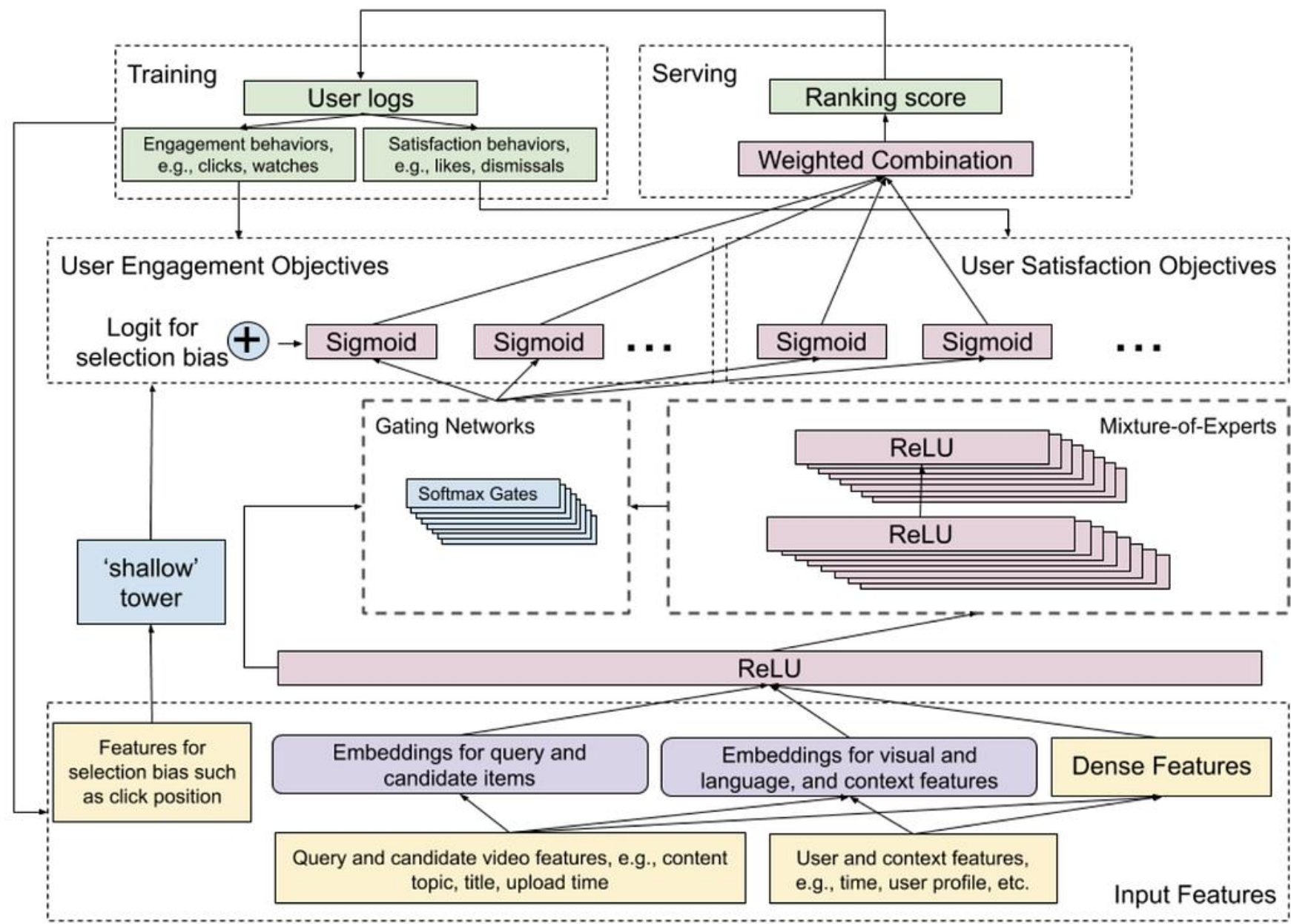


OFFLINE

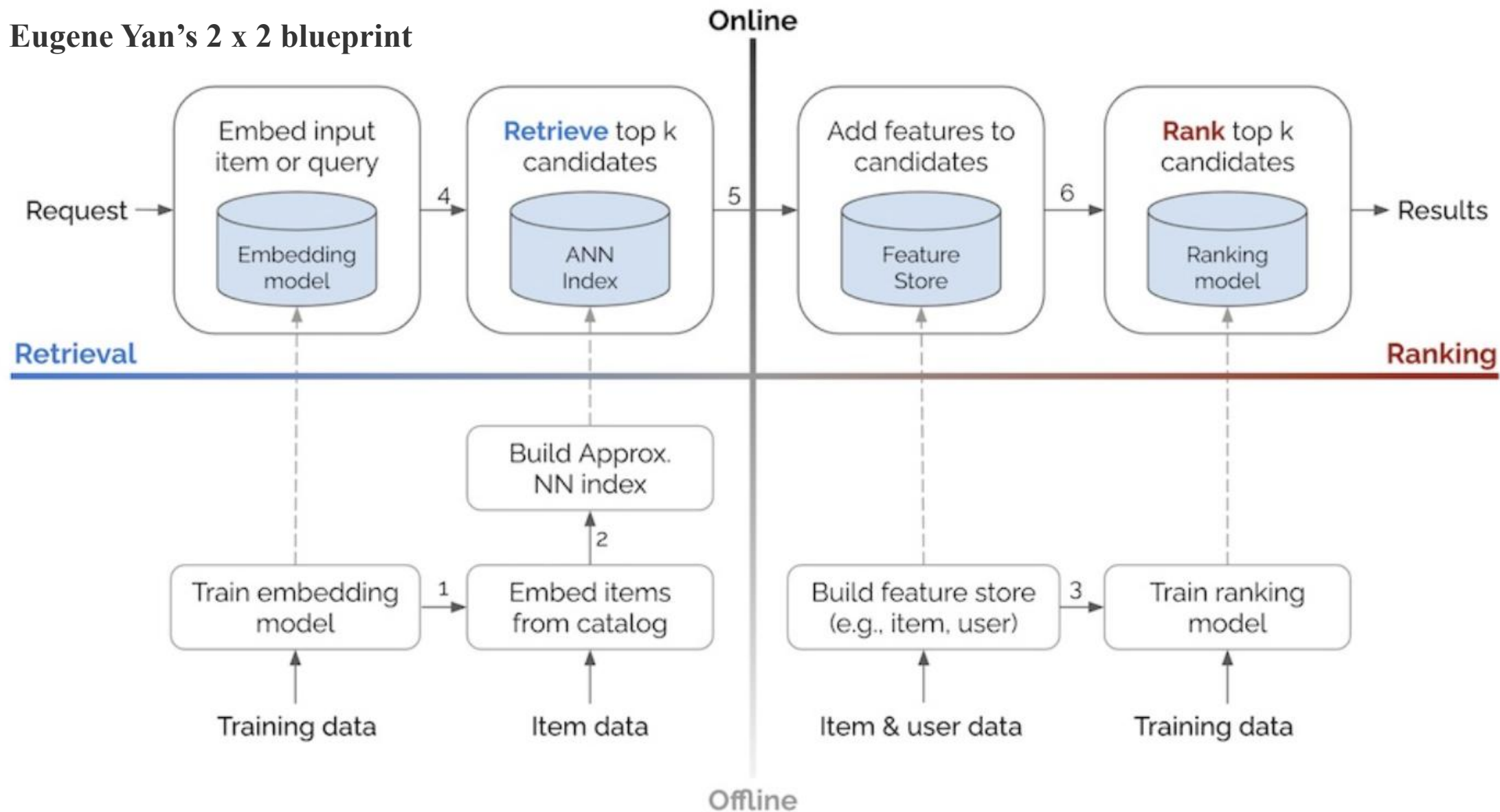
NEARLINE

ONLINE

YouTube Recommendation Platform



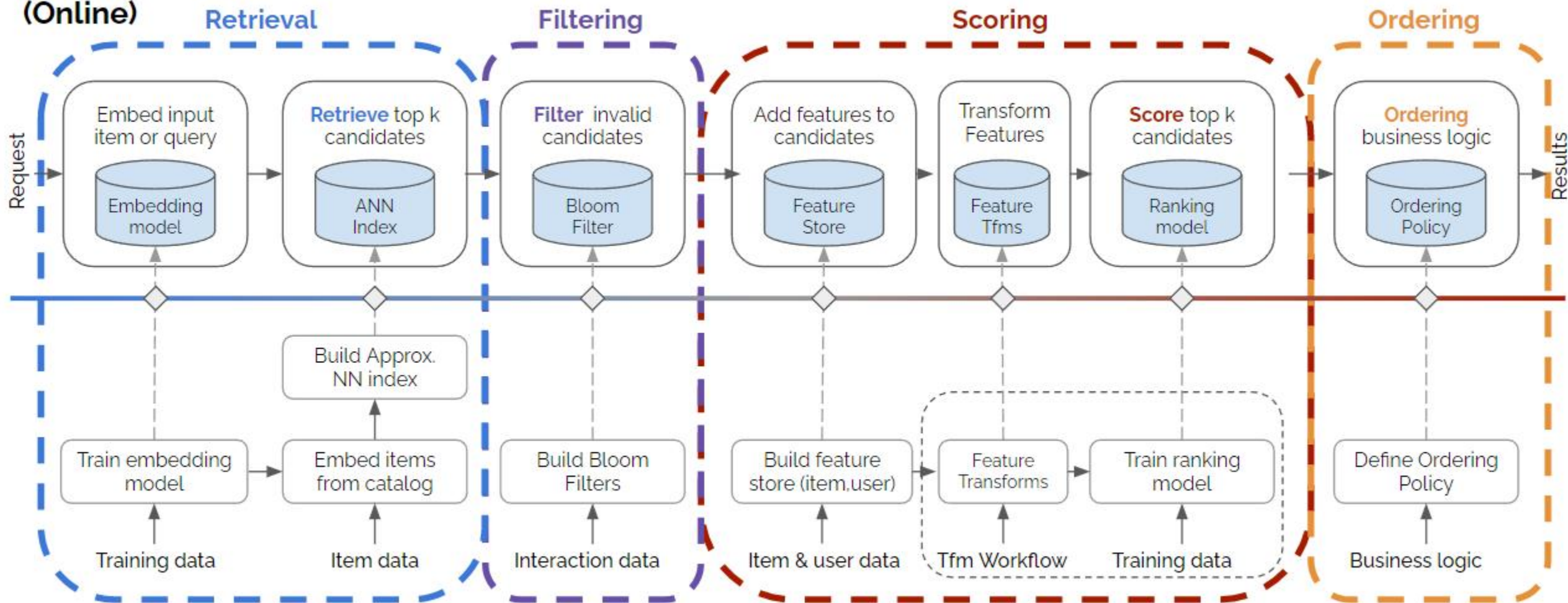
Eugene Yan's 2 x 2 blueprint



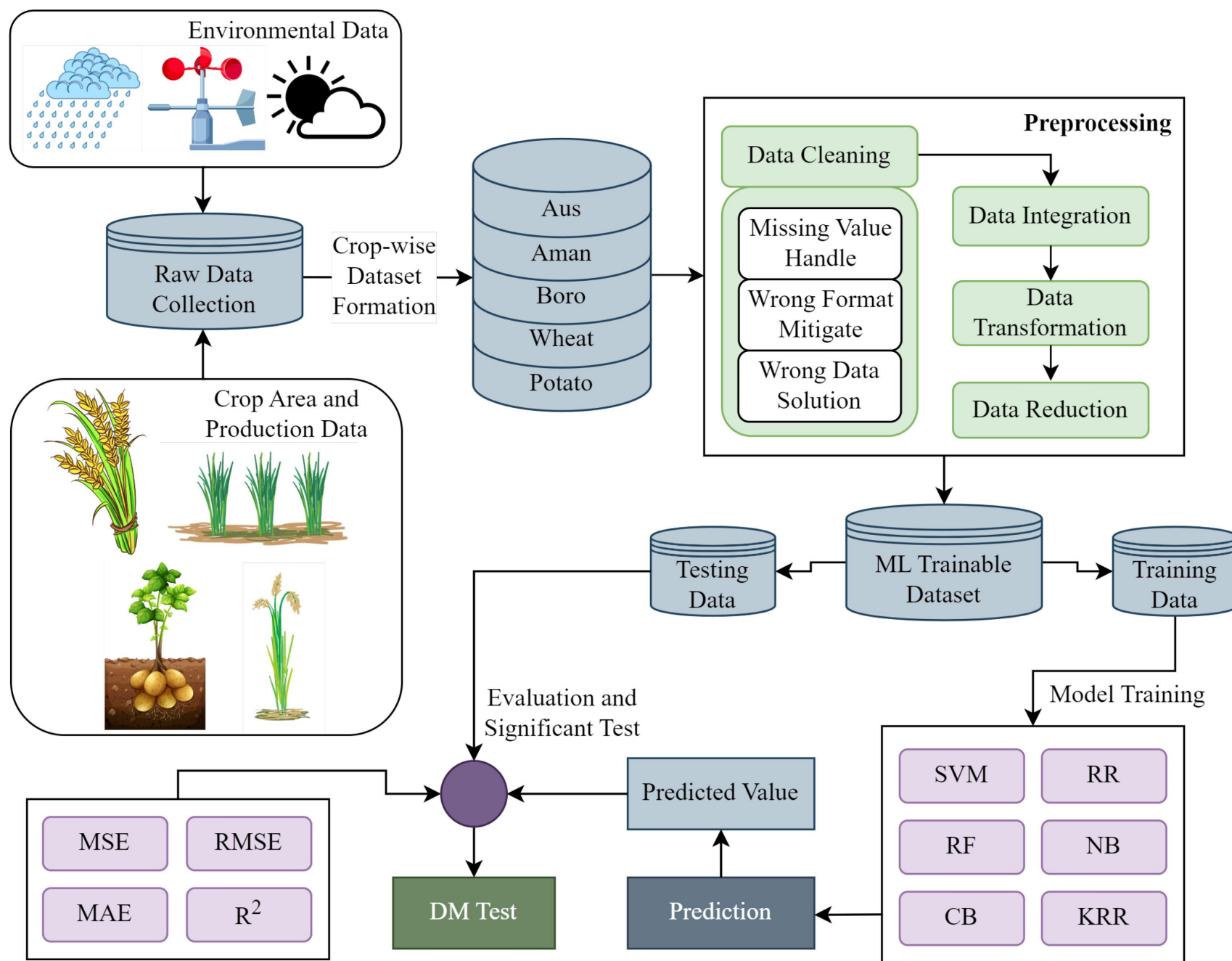
Basic system design for recommendations and search, based on the 2 x 2 above.

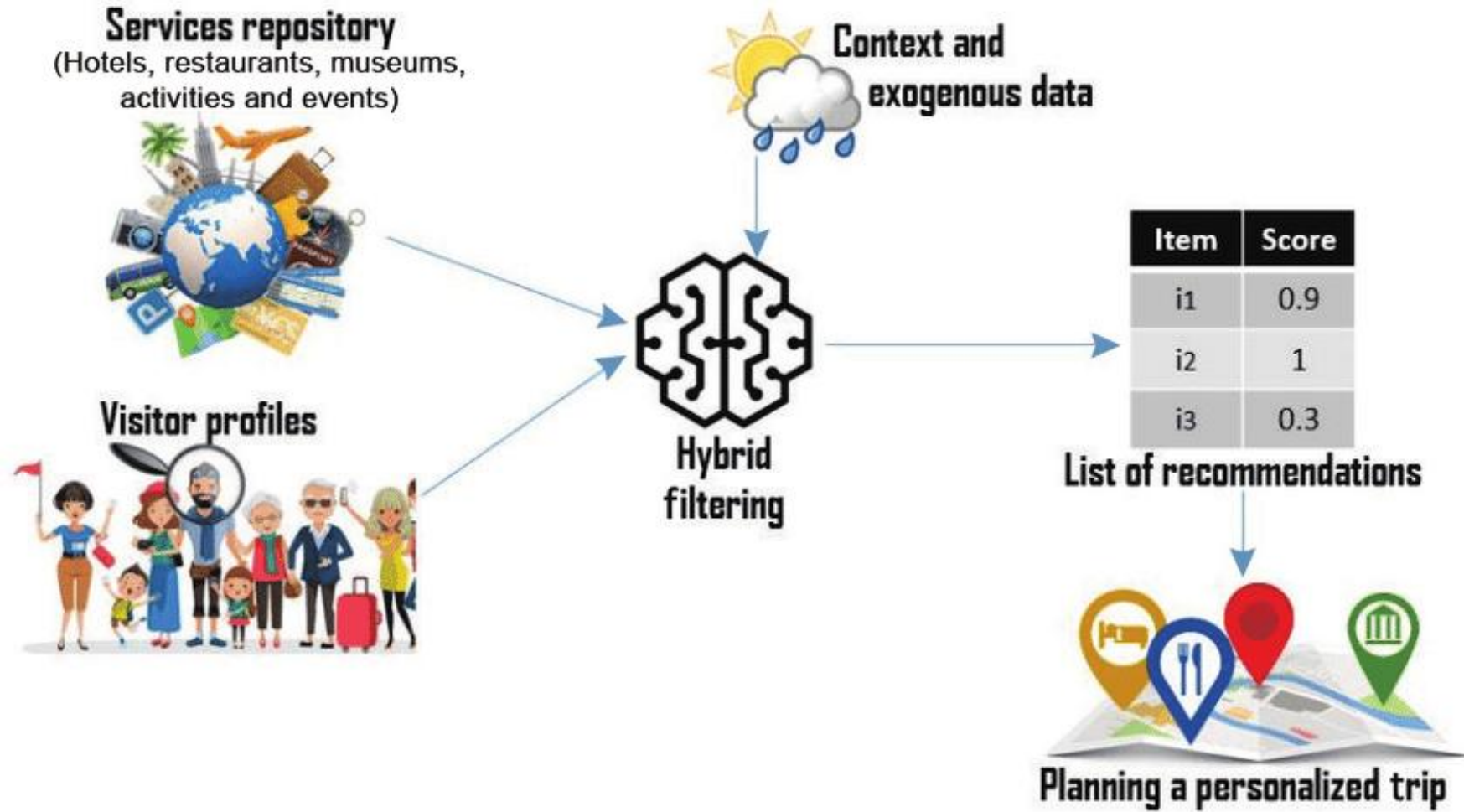
Nvidia's 4 stage blueprint

Inference (Online)



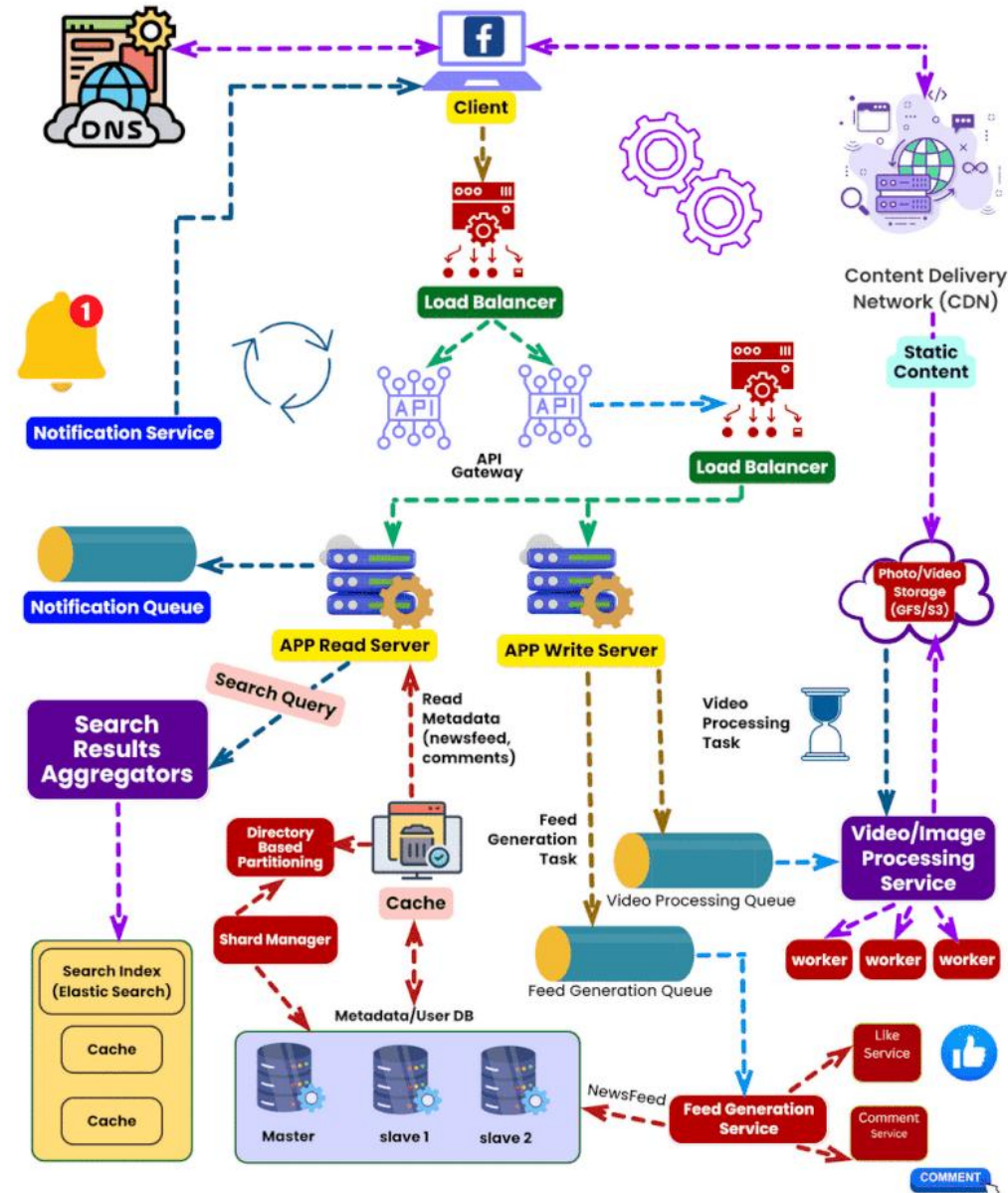
Training (Offline)

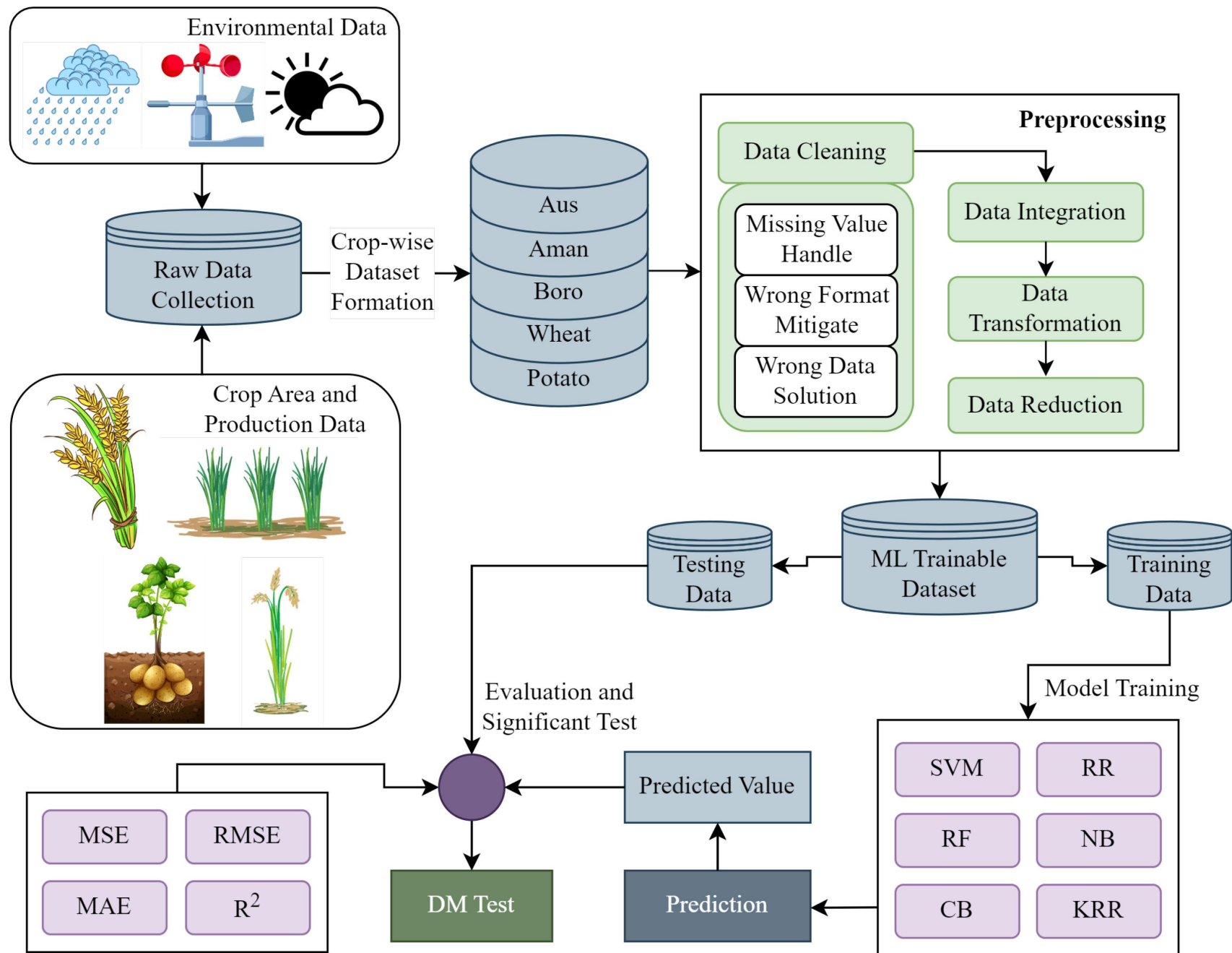


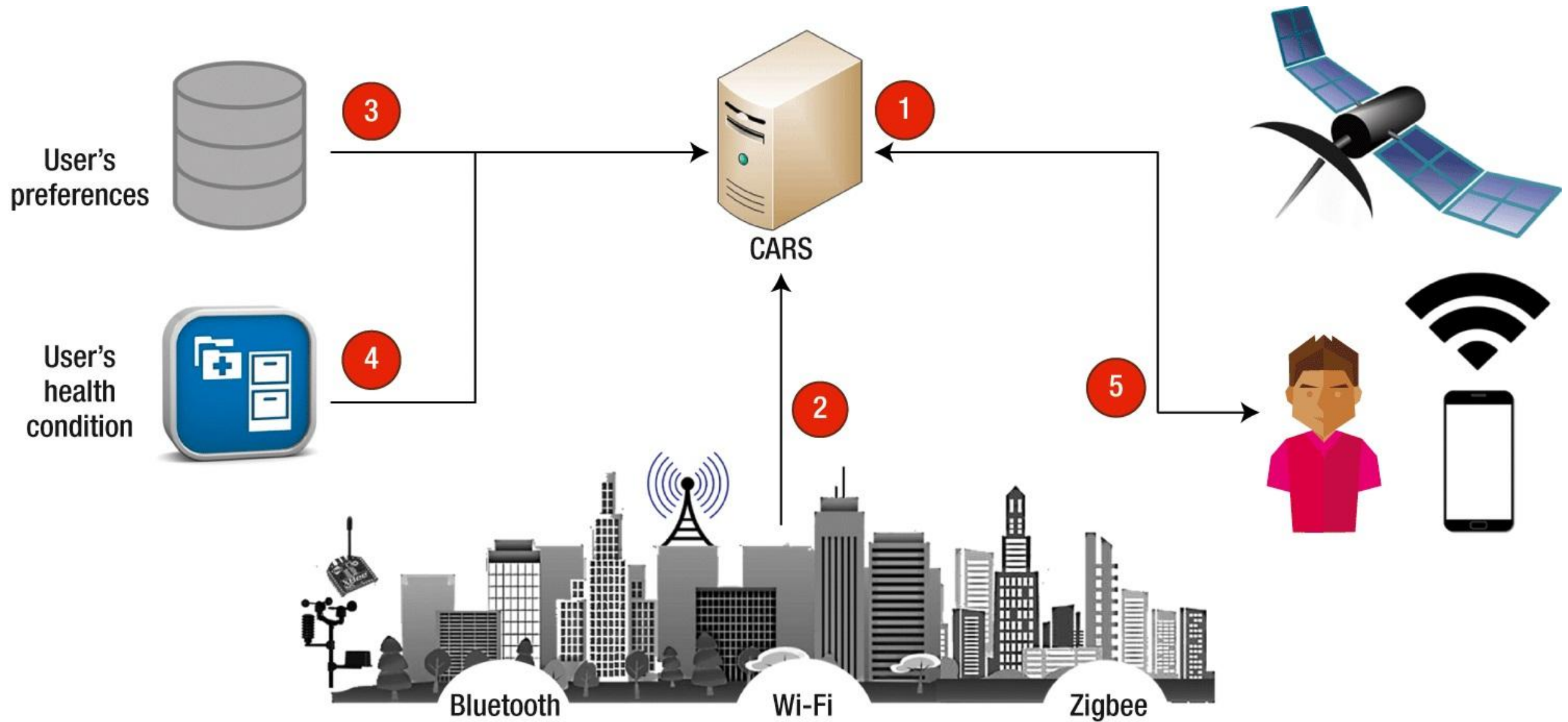


Facebook System Design

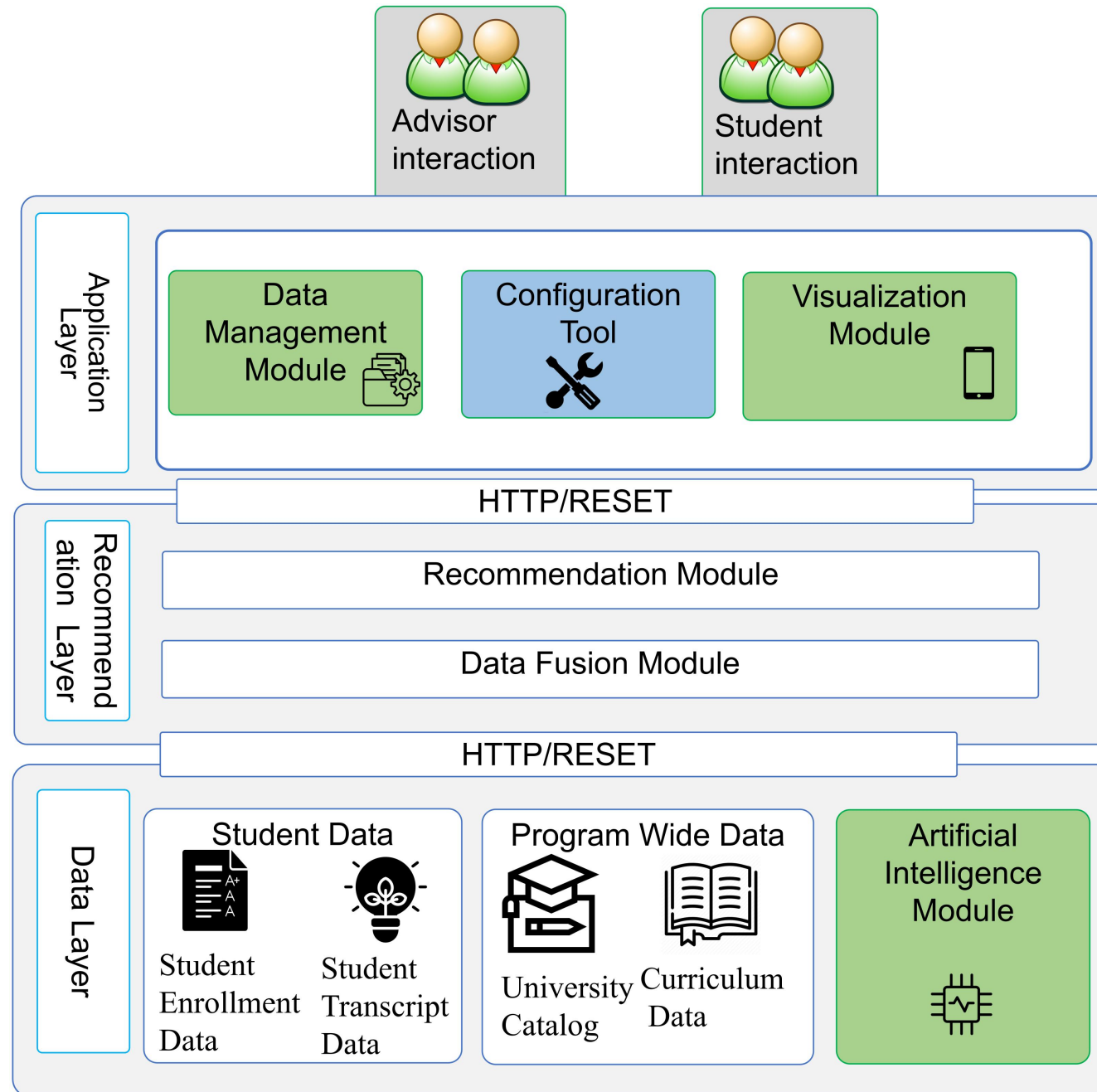
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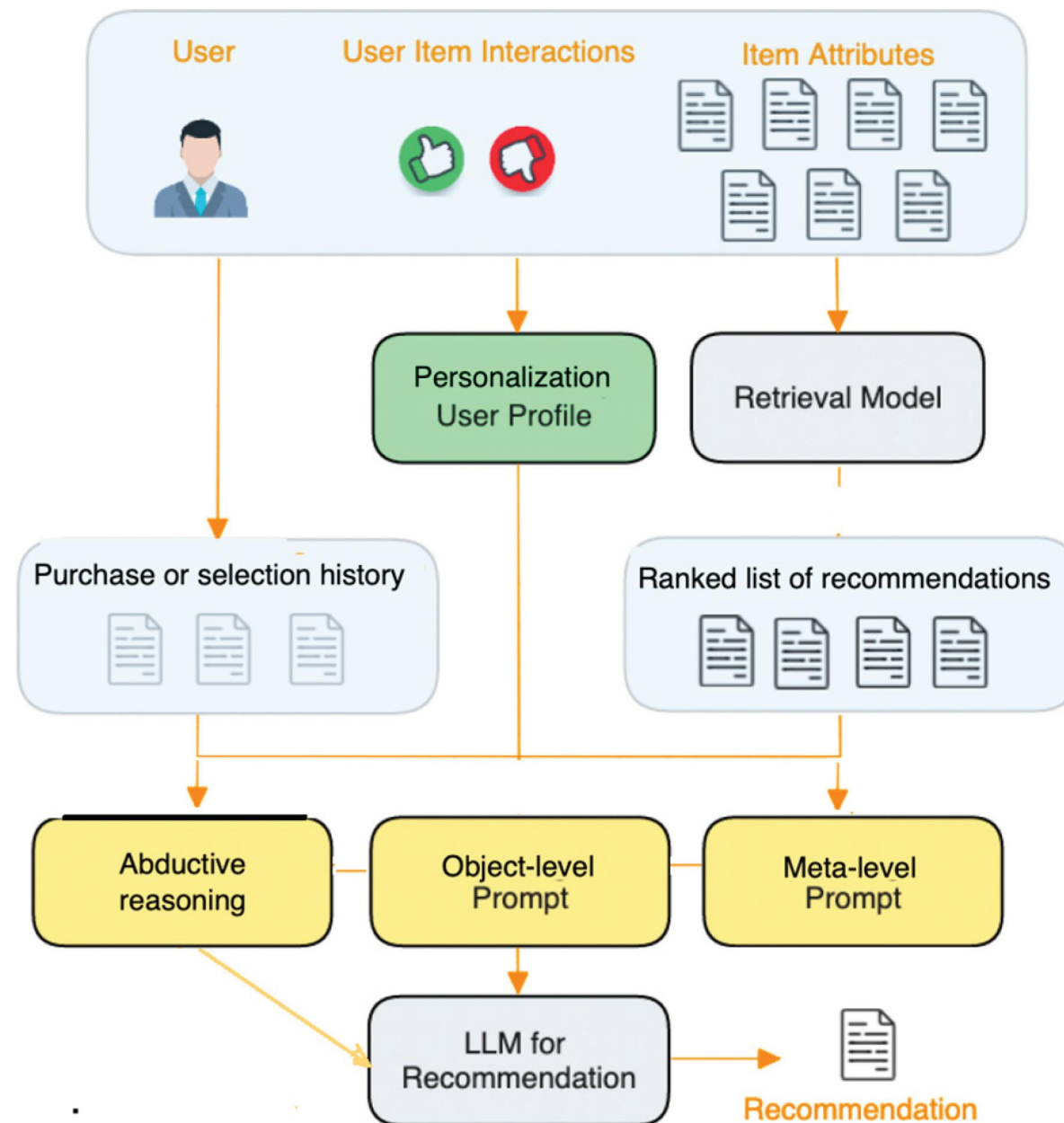






Smart City Fitness Recommender Systems:

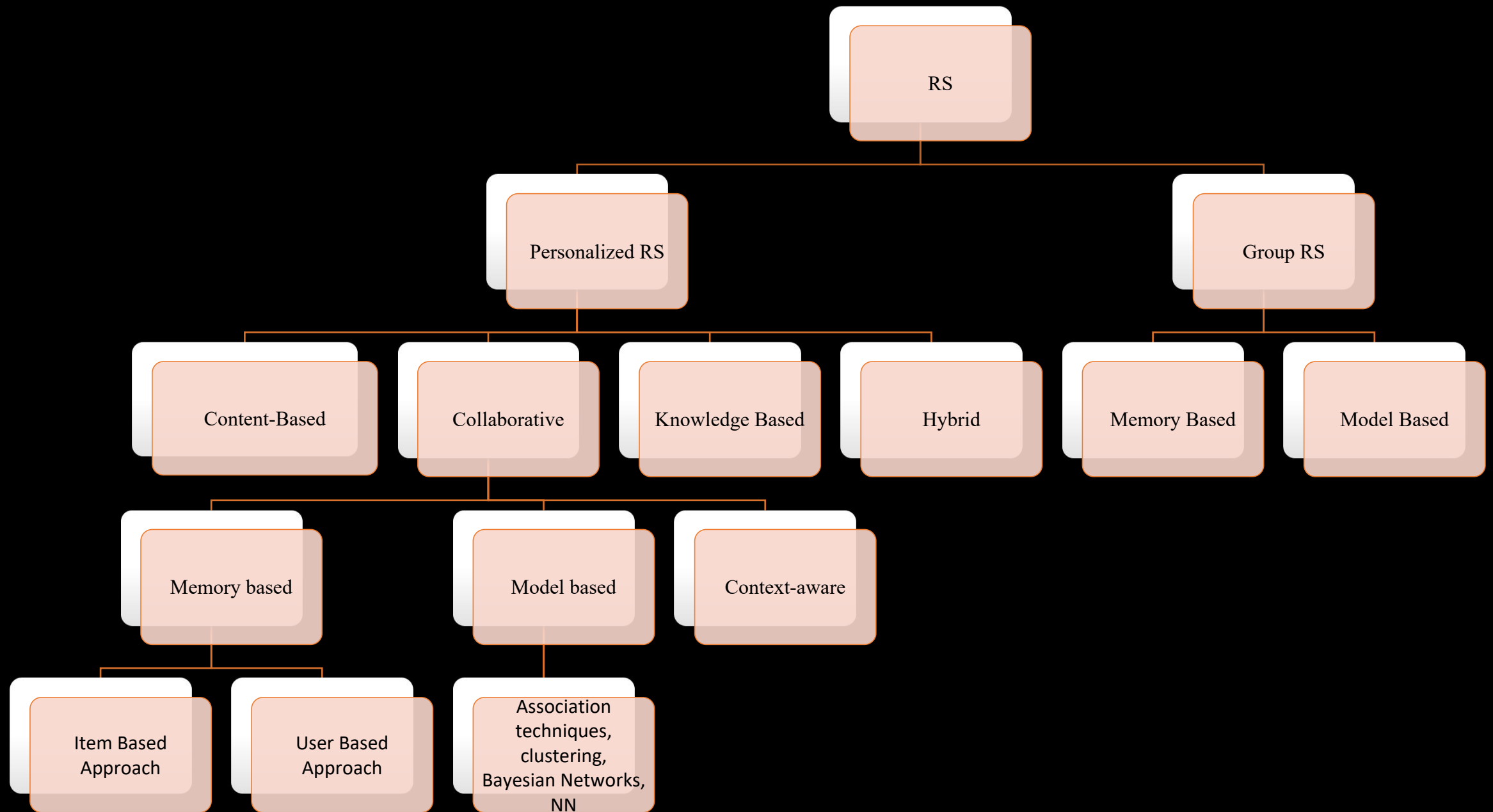




LLM- Based Personalized Recommendations in Health

Challenges in RS – Commercial Perspective

- **Robustness** - Robustness is the ability of a recommender system to maintain accurate and effective recommendations in the face of varying or unexpected data.
- **Fairness** - Fairness denotes the principle that a recommender system should provide unbiased and equitable recommendations for all users, regardless of their demographic or personal characteristics
- **Data Bias** - Data bias is the systematic deviations in the data used by a recommender system that can result in unfair or inaccurate recommendations.



Collaborative Filtering (CF)

- A **collaborative filtering-based recommender system** is one of the most widely used approaches and it makes recommendations based on the behavior of other users.
- This type of recommender system can be broadly classified into three categories: **memory-based**, **model-based**, and **context-aware based**.
- This method of recommendation is driven by the **user profiles** that are created from their **historical interactions**, such as their **past browsing history**, the items they have purchased, or the songs they have listened to.

Collaborative Filtering (CF)

- Collaborative Filtering-based (CF) methods are primarily used in **big data** processing platforms due to their **parallelization characteristics**.
- CF recommendation systems use the behavior of a group of users to recommend to other users.

CF...

- **Memory-based CF-**



Problem/ Issues

- It often encounters challenges in the form of **sparsity in the interaction matrix** and **computational difficulties** in large user or item spaces.
- The calculation of similarity between users or items is a crucial aspect of recommender systems.
- In collaborative filtering, the similarity is primarily obtained from the co-rated items and is used to predict the preference of an item or user.

CF...

- **Memory-based CF-**
- There are various methods to calculate the similarity, such as **Pearson Correlation Coefficient (PCC)**, **Cosine similarity (COS)**, **Mean Squared Difference (MSD)**, **Proximity Impact Popularity (PIP)**, **Jaccard Similarity**, and **Proximity Significance Singularity (PSS)**
- To reduce the **computational complexity** in recommender systems, the use of embedding techniques has been widely adopted.
- For instance, Valcarece et al. proposed the “prefs2vec” model, which is inspired by the bagof-words (CBOW) in “word2vec” to learn the user and item embedding.
- On the other hand, Chen et al. applied a graphbased method to learn the user and item representation.

CF...

- **Memory-based CF-**
- Memory-based CF is a widely used recommendation technique that leverages the **similarity between users or items** to make recommendations.
- The algorithm is comprised of two key steps: **first**, the similarity between users or items is calculated using a nearest neighbor-based method; and **second**, recommendations are generated for users based on the sorted similarity values.
- The ability to capture **user behavior patterns** and preferences based on **historical data**, as well as its ability to handle large amounts of data, makes memory-based CF a valuable tool for personalized recommendation.

CF...

- Model-based CF-
- Model-based CF is another widely adopted recommendation method that **predicts** the preferences of users based on the relationships between them and the items.
- **In recent years, there has been a growing interest in the application of neural networks in recommendation systems.**
- The integration of review text data has been shown to improve the performance of recommendation systems. This is why models like Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), Recurrent Neural Network (RNN), and Auto-Encoder have been widely adopted in recent studies [12]–[14], and they can be named neural-based CF in general.

CF...

- Model-based CF-

- One of the approaches is utilizing Multi-Layer Perceptron (MLP) for latent feature learning [15], resulting in successful models such as Neural Collaborative Filtering (NCF) [16] and Neural Network Matrix Factorization (NNMF) [17].
- These models provide personalized recommendations to users and have proven to be highly effective [18].
- The basic framework of the neural-based CF consists of mapping user and item representations, s_u and s_i , respectively, as well as reviewing text data, t_{ui} , into a continuous vector space through a neural network NN;

$$\hat{r} = NN(s_u, s_i, t_{ui} | \theta) \quad (1)$$

- The model parameters θ are then optimized to minimize the prediction error. The output of the neural network, $\hat{r}$, represents the predicted rating and is usually fed into a factorization-based method for further refinement.

CF...

- **Context Aware CF-** To enhance the decision-making capabilities of the recommendation system (RS), it is important to consider contextual information such as time, location, social status, mood, weather, day type, language, etc.
- The relationship between user activity and contextual information is closely intertwined, as they both have a reciprocal effect on each other.
- This is why incorporating contextual information into the model results in a better recommendation.
- **Sources of contextual information..????**
- Context information from smart devices is more complex, including, but not limited to, accelerometers, gyroscopes, magnetometers, GPS, biometric sensors, and so on, which can number up to 600 variables. Therefore, considering all this contextual information in a recommender system is a challenging task.
- To reduce its complexity, a direct approach is to use latent context embedding to reduce the dimension. – **Autoencoders..**

CF...

- Context Awareness
- consider contextual information
- The relationship between user and item is important to each other.
- This is why incorporating context information is important
- Sources of contextual information
- Context information from user, item, environment, biometric sensors, and system is a challenging task
- To reduce its complexity

SUMMARIZATION OF CONTEXTUAL AWARE RECOMMENDER SYSTEMS.

Application	Contexts dimension	Method	Model Applied
event and movie recommendation [52]	some explicit contextual features	Contextual modeling	denoising autoencoder neural network
food, mobile app recommendation [44]	hungry status (hungry, normal and full), daytime, day of the week, and location	Pre-filtering	attentive interaction network
POI, movie, and music recommendation [46]	time, location, ringer mode, battery, activity recognition, light, accelerometer, orientation, etc.	Contextual modeling	matrix factorization
mobile app, music and movie recommendation [53]	user ID, item ID, daytime, the last music ID that the user has listened within 90 minutes, movie ID and movie genres (multi-hot), etc.	Contextual modeling	convolution neural network
POI and movie recommendation [50]	time, location, companion, and travel type (family, couples, business, solo travel, friends)	Pre-filtering	neural collaborative filtering
POI and mobile app recommendation [51]	location, noise level, part of the week, time of day, day of the week, etc.	Contextual modeling	neural matrix factorization
e-commerce [8]	time, time difference, event type (view, sale or add to basket), etc.	Contextual modeling	graph convolution, encoder, decoder

important to

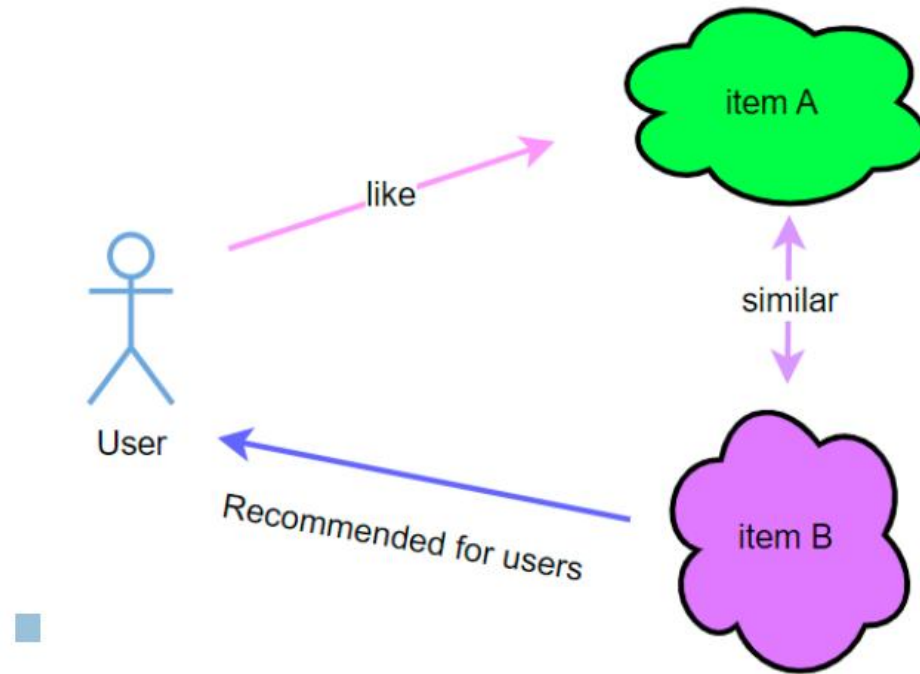
1 each other.

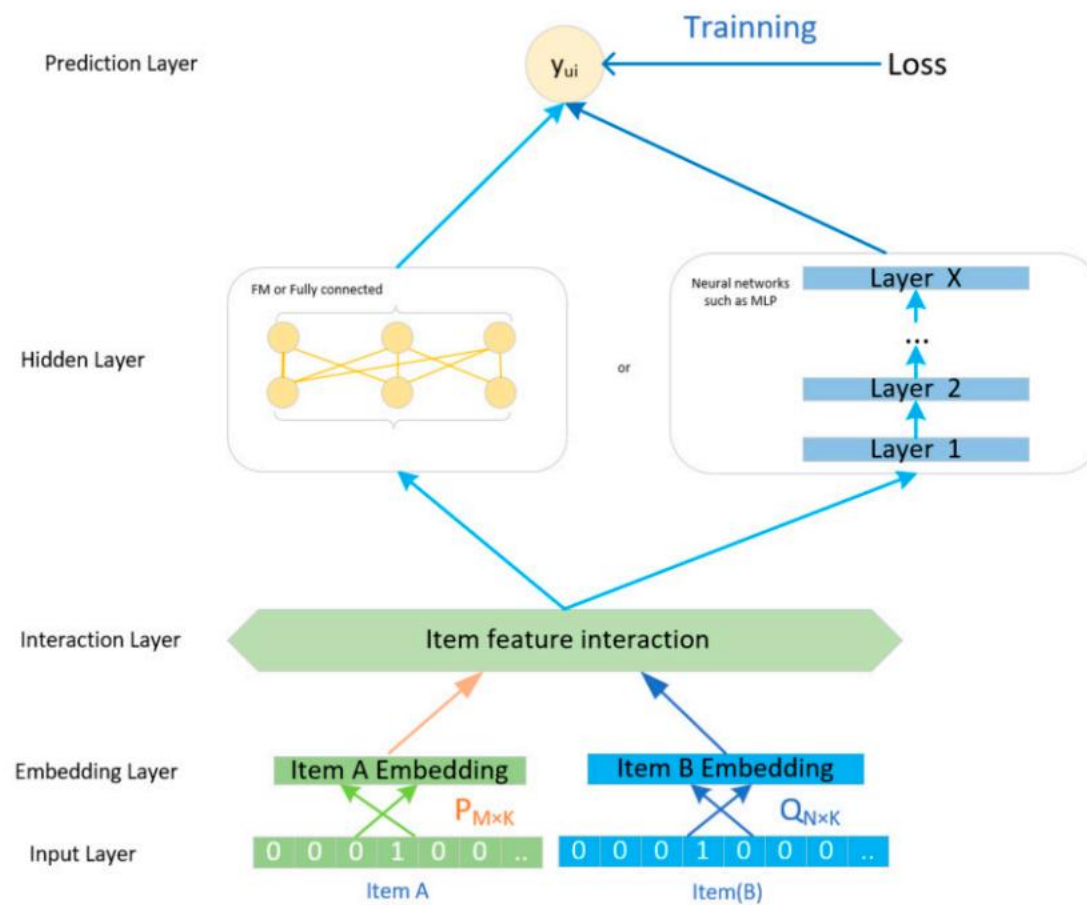
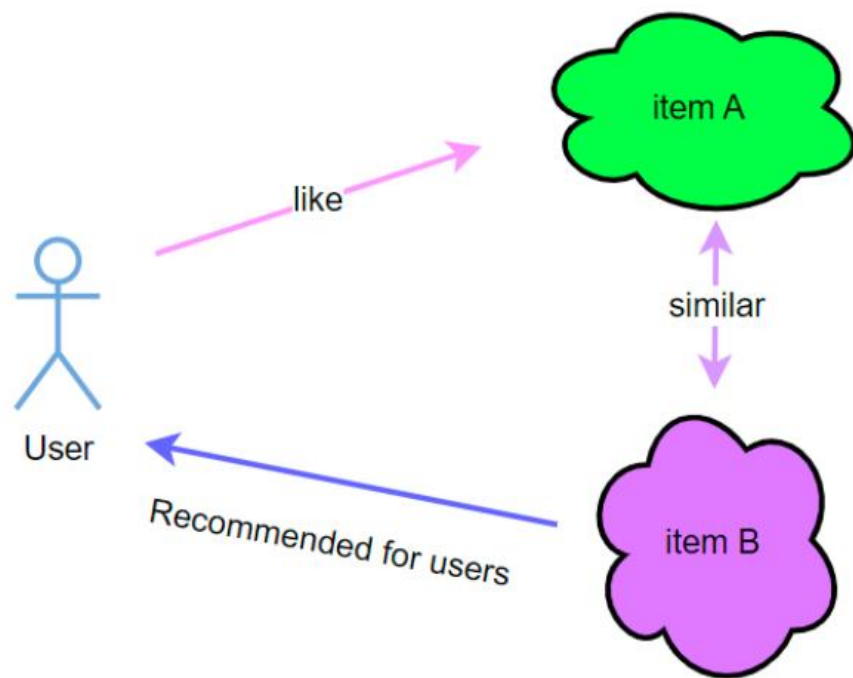
eters, GPS, recommender

Content-based Filtering (CB)

- Unlike collaborative filtering, content-based RS treats recommendations as a **user-specific** classification problem by learning a classifier based on item features, and it provides a more personal recommendation.
- The system mainly focuses on two types of information which are: User preference model; and User interaction history.
- The main idea of content-based recommenders is to recommend items based on the similarity between different users or items [20].
- This algorithm determines and differentiates the main common attributes of a particular user's favorite items by analyzing the descriptions of those items.
- In content-based RS, the Heterogeneous Information Network (HIN) approach is widely used to connect different types of side information in a parallel manner, as many recommender systems involve multiple item types and related links.
- HIN is a unique network that incorporates multiple item types and relationships to effectively integrate data from one or more domains, adding richer semantics to items and links. These networks can take the form of social networks, knowledge graphs, or other types of information networks.
- In the recommendation process, either **clustering or ranking based** methods can be used. Typically, clustering is based on the Bayesian model, while the ranking is based on the distance between candidates' and users' preferences. Different similarity metrics, including Kendall-Tau, Spearman, Pearson, etc., can be used for similarity calculation.

Content-based Filtering (CB)





NN Based

The issue of sequential recommendation entails user-based learning to anticipate user interactions with items. The aim of sequential recommendation is to regularly model users' sequential behaviors. We delve into the intricacies of recurrent-based recommendation models. To offer a more intuitive understanding of this concept, Figure 4 presents the architectural diagram of a typical sequential-based recommendation model.

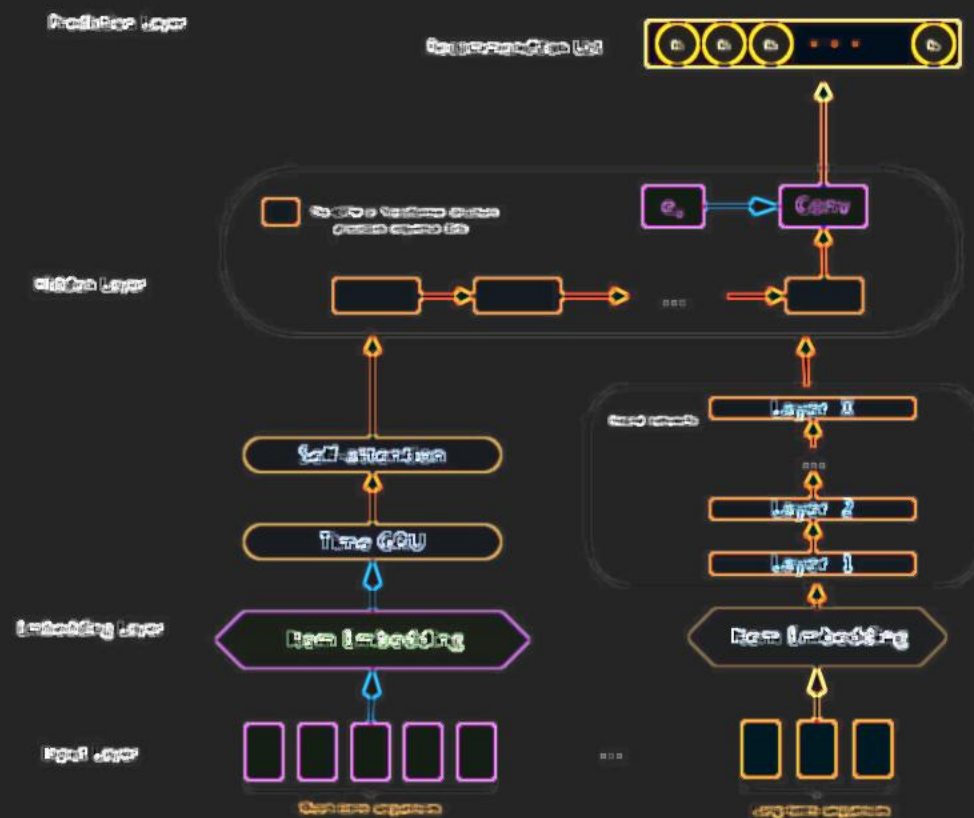
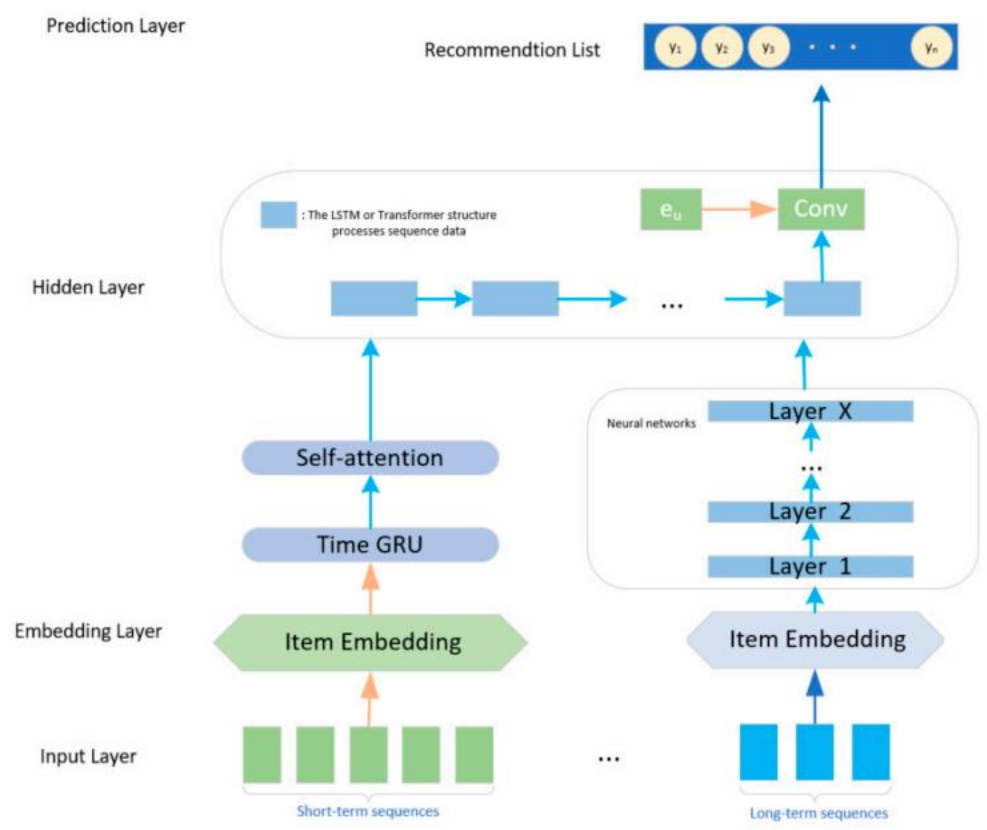


Figure 4. Sequential recommendation procedure

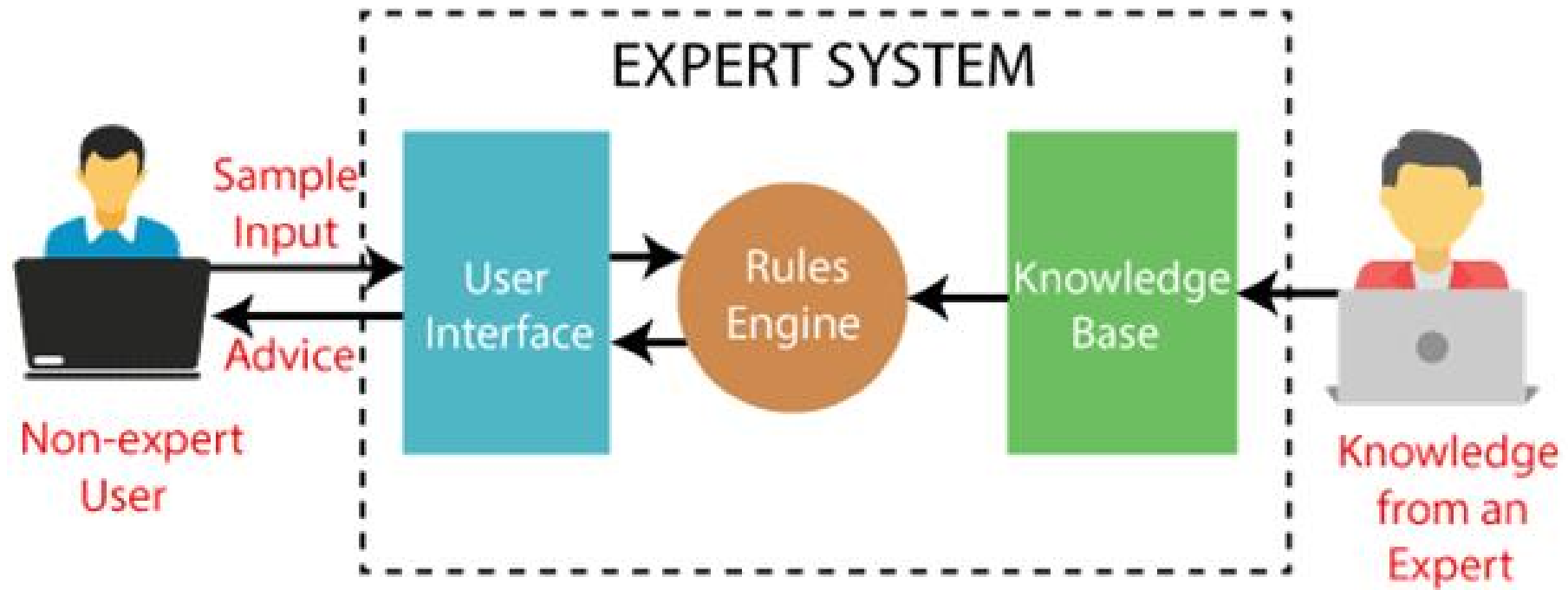
As illustrated in Figure 4, recurrent-based recommendation models are capable of continuously learning from user behavior. This sequential recommendation model

PDF sidebar showing a table of contents with page numbers 5, 6, 7, 8, and 9. Page 9 is currently selected and highlighted.



Knowledge-based RS

- The knowledge-based RS operates differently from CF based and content-based RS as it does not rely on user item interaction history.
- It is commonly used in complex domains where items are not typically purchased, such as cars and department businesses.
- The key information for knowledge-based RS is domain knowledge, which can encompass user knowledge, item knowledge, or the interaction between users and item.
- Huang et al. proposed the **Knowledge-aware Coupled Graph Neural Network (KCGN)**, which incorporates inter-dependent knowledge across items and users into the recommendation framework.
- This enhances the encoding of high-order user- and item-wise relationships, as well as the utilization of mutual information for global graph structure awareness



Knowledge-based RS – Generative AI

- Wang et al. explored the intents behind user item interactions using auxiliary item knowledge, proposing the **Knowledge Graph-based Intent Network (KGIN)**. This model models each intent as an attentive combination of KG relations, promoting independence among different intents for better model capability and interpretability.
- Xia et al. introduced the **Knowledge-Enhanced Hierarchical Graph Transformer Network (KHGT)** to explore multi-typed interactive patterns between users and items in recommender systems. The model is based on a graph-structured neural architecture, which focuses on capturing type-specific behavior semantics, as well as explicitly discriminating the importance of different types of user-item interactions.
- In the medical sector, Gong et al. proposed a novel framework, **Safe Medicine Recommendation (SMR)**, that builds upon recent advances in graph embedding learning techniques.
- **SMR first creates a high-quality heterogeneous graph by connecting Electronic Medical Records (EMRs) and medical knowledge graphs, and then jointly embeds diseases, medicines, patients, and their corresponding relationships into a shared lower-dimensional space**

Hybrid RS

- The development of recommender systems has resulted in increased complexity, particularly in their practical applications.
- To achieve real-time, accurate recommendations, these systems often require the integration of various data sources.
- The use of deep learning has allowed for the more effective fusion of multi-source information, such as context and content, in hybrid recommender systems.
- Polignano et al. blend user-item interactions and contextual features for the recommendation. Luo et al. mixed graph embedding and context information for the recommendation

Hybrid RS

THE SUMMARY ABOUT THE HYBRID RECOMMENDER SYSTEM BASED ON THE MODEL AND DATA SOURCE.

Core Model	Composition	Information Source	Application
Graph Model (LCP theory) [116]	Content based, CF based	Content (Papers), Context (Authors' social network)	Researcher Recommendation
Graph Model [117]	Content-based, CF based	Content (Music Descriptions), Context (User Preferences)	Music Recommendation
Hybrid Probabilistic Graphical Model [118]	Content-based, CF based	Content (Music Style), Context(User's history, social connections, tags, and popularity statistics)	Music Recommendation
AutoEncoder [119] [120] [121]	CF based	Content (User, Item), Context (User's ID, age, gender, occupation, zip code, movie genres, title)	General Recommendation
DNN [113]	CF based	Content (User, Item), Context(Movie genres, title, user tags)	General Recommendation
SVD [122]	Content-based, CF based, Fuzzy Expert System	Content (User, Movie), Context(Average movie rating, number of ratings)	Movie Recommendation
Alternating Least Squares, Bayesian Personalized Ranking [123]	Content-based, CF based	Content (Authors from research articles, Chemical Compounds in ChEBI, Number of articles the author wrote)	Chemical Compound Recommendation
Alternating Least Squares, Collaborative Filtering, DNN [124]	Content-based, CF based	Content (User Preferences, Browsing/Search/clicks/ order history)	Business Recommendation

Group RS

- The field of recommendation systems has seen a significant amount of research in recent years, with the majority of work focusing on individual recommendation systems.
- However, there is a growing body of literature that deals with group recommendation systems, which aim to provide recommendations to a group of users who have similar preferences, as opposed to individualized recommendations.

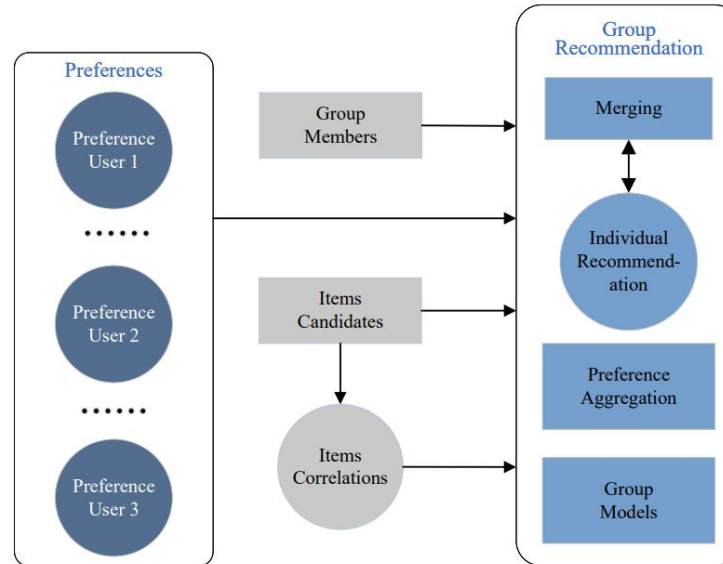


Fig. 2. A general framework of group recommender system. (The figure is from [125].)

Activity...

Explore Memory-based and Model-based
Group RS

How to measure performance of a RS ?

Performance Measures

- Generally, due to the application type and prediction target, they can be categorized into two parts:

Rating Based Indicator (RBI)

RBI means that the output of the recommender system is a rating score.

Item Based Indicator (IBI)

IBI means the output of the recommender system is a set or a list of items.

Rating Based Indicator (RBI)

- The rating-based indicator is to evaluate the quality of the prediction rating score.
- The direct way is to calculate the gap between **implicit/explicit labels**.
- One most popular measurements are Root Mean Squared Error (RMSE) when the rating score is explicit value.

$$RMSE = \sqrt{\frac{1}{|U||I|} \sum_{u \in U, i \in I} (\hat{r}_{ui} - r_{ui})^2}$$

where U is the set of the users, I is the set of items, and $\hat{r}$ denotes the predicted rating, and r denotes the true rating.

Rating Based Indicator (RBI)

- Mean Absolute Error (MAE) is another popular measurement

$$MAE = \sqrt{\frac{1}{|U||I|} \sum_{u \in U, i \in I} |\hat{r}_{ui} - r_{ui}|}$$

RMSE and MAE are non-negative, a lower RMSE and MAE are better than a higher one.

While each value $(\hat{r}_{ui} - r_{ui})^2$ in RMSE, $|\hat{r}_{ui} - r_{ui}|$ in MAE is proportional to the final error, which leads RMSE and MAE are sensitive to outliers.

Item Based Indicator (IBI)

- As the recommender system is a set or list of items, if there is no ranking information, the confusion matrix listed in Table V can be adopted in the evaluation.

THE CONFUSION MATRIX FOR THE RECOMMENDER SYSTEM.

	Recommended	Not Recommended
Used	True Positive (TP)	False Negative (FN)
Not Used	False Positive (FP)	True Negative (TN)

TP is the outcome of the used items that are recommended by the system,

FP is the outcome of not used items that aren't recommended by the system.

FN denotes the outcome of the used items that are not recommended by the system

TN is the outcome of the not used items that are not recommended by the system

Item Based Indicator (IBI)

- Generally, a more comprehensive composition of those four values is adopted in the recommender system, such as Precision, Recall, and F – Measure

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F\text{-Measure} = \frac{2 * Precision * Recall}{Precision + Recall}$$

Item Based Indicator (IBI)

Ranking Indicators: In fact, most recommendations are ordered lists of items, but Precision and Recall have nothing to do with order.

- Therefore, it makes more sense to take top-k P recision and Recall at cutoff k (P recision@k and Recall@k for short), which considers only a subset of the ordered list from the 1st item to the k-th item.
- A more macro indicator is to average all the users to obtain the mean average precision (MAP@k) and mean average recall (MAR@k),

$$MAP@k = \frac{\sum_{i=1}^k (Precision@i \times rel(i))}{\min(k, \text{number of relevant items})}$$
$$MAR@k = \frac{\sum_{i=1}^k (Recall@i \times rel(i))}{\text{total number of relevant items}}$$

where $rel(i)$ is an indicator function, if the i -th item is relevant to the target, it is equal to 1, otherwise is 0.

MAP@K

MAP@K measures the average precision of recommendations up to position K across all users.

It rewards systems that place relevant items higher in the recommendation list.

Calculation Steps:

1. Precision at position k: For each position k in the recommendation list up to K, calculate the precision if the item at position k is relevant.
1. Average Precision (AP) for a user: Average the precisions at positions where relevant items occur.
2. Mean Average Precision (MAP@K): Compute the mean of the Average Precisions across all users.

$$\text{MAP@K} = \frac{1}{U} \sum_{u=1}^U \text{AP@K}_u$$

MAR@K

MAR@K measures the average recall of recommendations up to position K across all users.

It assesses how well the system retrieves all relevant items within the top K recommendations.

Calculation Steps:

1. Recall at position k: For each position k in the recommendation list up to K, calculate the recall if the item at position k is relevant.
2. Average Recall (AR) for a user: Average the recalls at positions where relevant items occur.
3. Mean Average Recall (MAR@K): Compute the mean of the Average Recalls across all users.

$$\text{MAR@K} = \frac{1}{U} \sum_{u=1}^U \text{AR@K}_u$$

MAP@K:

Focuses on the precision of relevant items in the top K recommendations, emphasizing their positions.

MAR@K:

Focuses on the recall of relevant items in the top K recommendations, assessing the system's ability to retrieve all relevant items.

Both metrics provide insights into the performance of recommender systems, with MAP@K emphasizing the ranking quality and MAR@K emphasizing the coverage of relevant items.

Item Based Indicator (IBI)

- With these two macro indicators, we can find out the general performance of the recommender system in the real application.
- However, these two indicators have some drawbacks, for example, they are not fit for fine-grained numerical ratings.
- They require a threshold to convert fine-grained numerical ratings to binary correlations, which can lead to bias and loss of fine-grained information.

Limitations of MAP@K

1.Highly Sensitive to the Choice of K

The metric's value can vary significantly depending on how many recommendations (K) are considered. A small K might overstate performance (if relevant items appear early), while a large K might mask poor ranking quality

2.Insensitive to the Absolute Number of Relevant Items

MAP@K doesn't account for how many relevant items exist in total. Two systems could show the same MAP yet have very different overall recall depending on the dataset's relevant-item count.

3.Binary Relevance Assumption

MAP assumes relevance is binary (relevant vs. not), which means graded preferences (e.g., rating 1–5 stars) must first be thresholded, potentially introducing bias.

4.Poor Fit for Non-Ranked Scenarios

It's tailored for ranked outputs. In cases where order is irrelevant (e.g., batch recommendations with no priority), MAP@K may not be meaningful.

5.Affected by Query Difficulty

In contexts where certain users have few or no relevant items, MAP scores can drop despite reasonable performance. Ignores Non-relevant Items

MAP focuses only on relevant items and their position. It doesn't penalize non-relevant items explicitly, so sometimes the broader listing quality can be misleading

Limitations of MAR@K

1. Order Insensitivity (Less Penalization for Ranking)

MAR@K emphasizes the presence of relevant items in the top K, regardless of exact positions. This can allow poor ordering to go unnoticed - meaning relevant items lower in the list still contribute equally.

2. Same Sensitivity to K Choice

Like MAP, the selection of K heavily influences MAR results. A small K might omit many relevant items entirely; a large K could dilute ranking relevance

3. Binary Relevance Only

MAR, like MAP, treats relevance as binary. Nuanced user preferences or graded ratings are lost unless converted to a binary format.

4. Potential Skew with Few Relevant Items

For users with very few relevant items, averaging recall across positions can produce volatile or misleading results, reducing interpretability.

Item Based Indicator (IBI)

- To solve the problem that exists in $\text{MAP}@k$ and $\text{MAR}@k$, and take the ranking into the consideration, another widely used metric is normalized discounted cumulative gain (**NDCG**).
- Like $\text{MAP}@k$ and $\text{MAR}@k$, the goal of NDCG is to put highly relevant items at the top of the recommendation list, but it improves the recommendation evaluation quality with fine-grained numerical ratings.

NDCG@K

- NDCG evaluates how well a ranked list places more relevant items higher.
- It uses a “discount” that reduces the credit for items at lower ranks, and it supports graded relevance (e.g., 0–3), not just binary.
- The idea: compute DCG@K, compute the ideal DCG@K (IDCG@K) for a perfect ranking, then normalize: $\text{NDCG@K} = \text{DCG@K} / \text{IDCG@K}$ (hence, it's in $[0,1]$)

For a list cut at K:

- DCG@K (linear gain): $\sum_{i=1}^K \frac{\text{rel}_i}{\log_2(i+1)}$
- DCG@K (exponential gain): $\sum_{i=1}^K \frac{2^{\text{rel}_i} - 1}{\log_2(i+1)}$

Then $\text{NDCG@K} = \text{DCG@K} / \text{IDCG@K}$, where IDCG@K is the DCG of the same labels sorted by decreasing relevance.

However, there are some disadvantages of using NDCG, for example, it cannot handle incomplete ratings, in this case, the missing value can be filled with mean/median values before it is passed to NDCG

Item Based Indicator (IBI)

- If there is no user's relative ranking information for two items, then we can tie them together into a single item. In the evaluation metrics mentioned above, except NDCG, there is no such relative information.
- Another metric that is suitable for such a situation is the normalized distance-based performance measurement (NDPM).

NDPM

- NDPM stands for Normalized Distance-based Performance Measure.
- It quantifies how well the system's ranking aligns with the user's preferences, accounting gracefully for ties.
- It was introduced by Yao (1995) and later applied to recommender evaluation, notably in evaluating collaborative filtering systems.

The formula:

Let:

- C^- = number of **contradictory preference pairs** — cases where the system says item $A > B$, but the user prefers $B > A$.
- C^u = number of **compatible preference pairs involving ties** — cases where user's ranking ties two items but system ranks one higher than the other.
- C = total number of strict preference pairs in the user's ranking (i.e., user actually prefers $A \neq B$).

Then:

$$\text{NDPM} = \frac{2C^- + C^u}{2C}$$

Because it's normalized, NDPM ranges from 0 (perfect agreement) to 1 (complete disagreement) and is comparable across datasets.

Content-based RS – State of the Art and Trends

CB-RS

- Recommender systems have the effect of guiding users in a personalized way to interesting objects in a large space of possible options.
- Content-based recommendation systems try to recommend items similar to those a given user has liked in the past

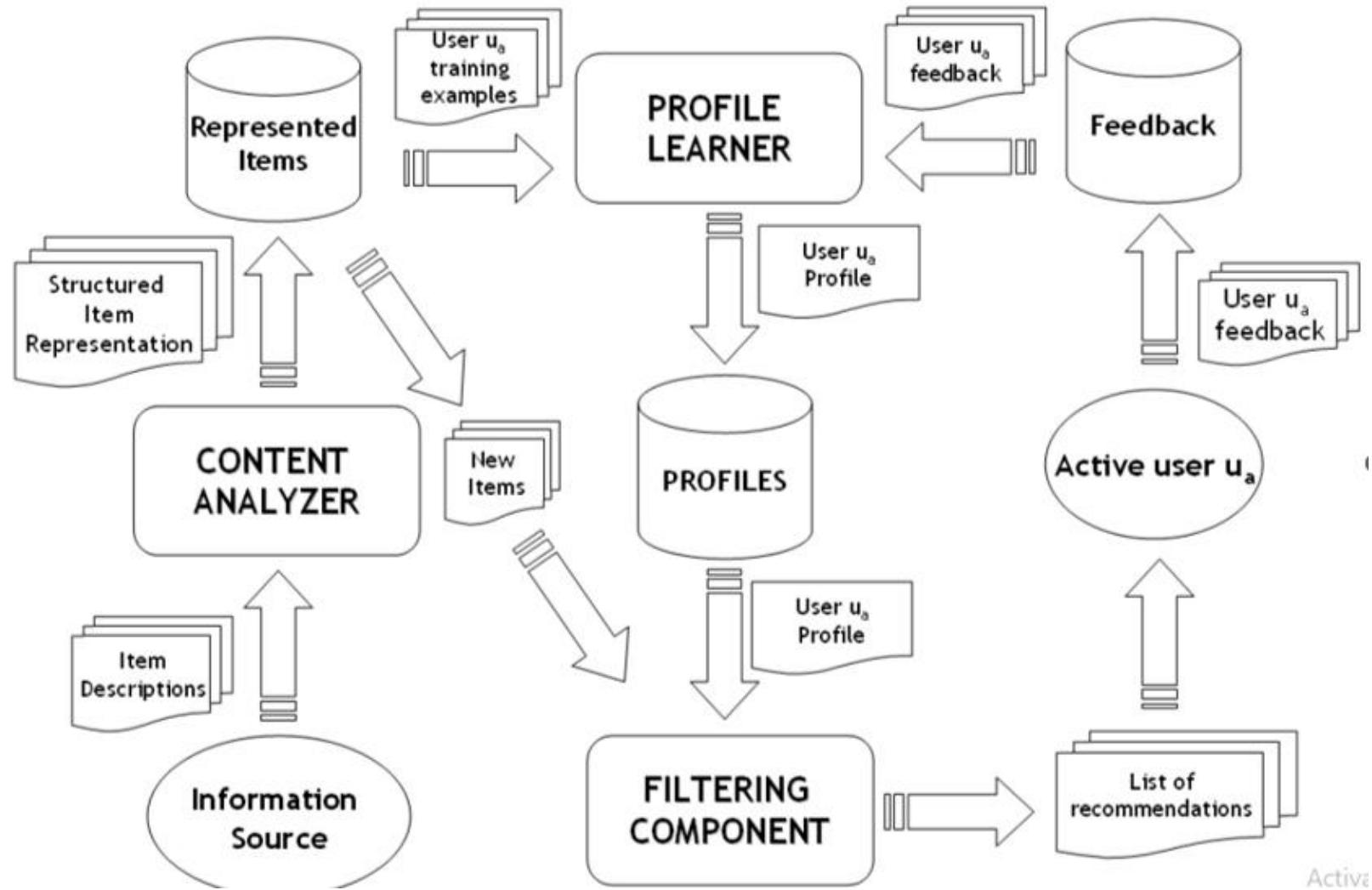


Fig. High level architecture of a Content-based Recommender

Item Representation

- Items that can be recommended to the user are represented by a set of features also called attributes or properties.
- When each item is described by the same set of attributes, and there is a known set of values the attributes may take, the item is represented by means of structured data. In this case, many **ML algorithms** can be used to learn a user profile.

Item Representation..

- In most content-based filtering systems, item descriptions are textual features extracted from **Web pages, emails, news articles or product descriptions**.
- Unlike structured data, there are no attributes with well-defined values.
- Textual features create a **number of complications** when learning a user profile, due to the natural language ambiguity.
- The **problem** is that **traditional keyword-based profiles** are unable to capture the semantics of user interests because they are primarily driven by a string matching operation.
- If a string, or some **morphological variant**, is found in both the profile and the document, a match is made, and the document is considered as relevant.

Item Representation..

- **Semantic analysis** and its integration in personalization models is one of the most innovative and interesting approaches proposed in literature to solve those problems.
- The key idea is the adoption of knowledge bases, such as **lexicons or ontologies** , for annotating items and representing profiles in order to obtain a “semantic” interpretation of the user information needs.

Item Representation..

- **Semantic analysis** and its integration in personalization is an innovative and interesting approaches for recommendation problems.

These formalize concepts, their properties, and interrelations—often used with description logics. Ontologies enable reasoning about "is-a" hierarchies, associations, and domain-specific knowledge. They provide deeper context beyond mere synonyms.

- The key idea is the adoption of knowledge bases, such as **lexicons or ontologies**, for annotating items and representing profiles in order to obtain a “semantic” interpretation of the user information needs.
- Semantic Lexicons (e.g., WordNet, EuroWordNet): These are structured dictionaries where words are grouped into synsets (synonym sets) and linked via semantic relations like hypernymy/hyponymy. They help in understanding term meaning, aiding tasks like query expansion and word sense disambiguation.

Keyword-based Vector Space Model

- Most content-based recommender systems use relatively simple retrieval models, such as **keyword matching** or the **Vector Space Model (VSM)** with basic **TF-IDF weighting**.
- VSM is a spatial representation of text documents.
- In that model, each document is represented by a vector in a **n-dimensional space**, where each dimension corresponds to a term from the overall vocabulary of a given document collection.

Keyword-based Vector Space Model

- Formally, every document is represented as a vector of term weights, where each weight indicates the degree of association between the document and the term.
- Let $D = \{d_1, d_2, \dots, d_N\}$ denote a set of documents or corpus, and $T = \{t_1, t_2, \dots, t_n\}$ be the dictionary, that is to say the set of words in the corpus.
- T is obtained by applying some standard natural language processing operations, such as tokenization, stopwords removal, and stemming.
- Each document d_j is represented as a vector in a n -dimensional vector space, so $d_j = \{w_{1j}, w_{2j}, \dots, w_{nj}\}$, where w_{kj} is the weight for term t_k in document d_j .

Keyword-based Vector Space Model

- Document representation in the VSM raises two issues: weighting the terms and measuring the feature vector similarity.
- The most commonly used term weighting scheme, **TF-IDF (Term Frequency-Inverse Document Frequency) weighting**, is based on empirical observations regarding text.
- rare terms are not less relevant than frequent terms (IDF assumption);
- multiple occurrences of a term in a document are not less relevant than single occurrences (TF assumption);
- long documents are not preferred to short documents (normalization assumption).

Keyword-based Vector Space Model

- In other words, terms that occur frequently in one document (TF = term-frequency), but rarely in the rest of the corpus (IDF = inverse-document-frequency), are more likely to be relevant to the topic of the document.
- In addition, normalizing the resulting weight vectors prevent longer documents from having a better chance of retrieval.
- These assumptions are well exemplified by the TF-IDF function

$$\text{TF-IDF}(t_k, d_j) = \underbrace{\text{TF}(t_k, d_j)}_{\text{TF}} \cdot \underbrace{\log \frac{N}{n_k}}_{\text{IDF}}$$

where N denotes the number of documents in the corpus, and n_k denotes the number of documents in the collection in which the term t_k occurs at least once.

Keyword-based Vector Space Model

$$\text{TF}(t_k, d_j) = \frac{f_{k,j}}{\max_z f_{z,j}}$$

- where the maximum is computed over the frequencies $f_{z,j}$ of all terms t_z that occur in document d .
- In order for the weights to fall in the $[0,1]$ interval and for the documents to be represented by vectors of equal length, weights obtained by Equation are usually normalized by cosine normal.

$$w_{k,j} = \frac{\text{TF-IDF}(t_k, d_j)}{\sqrt{\sum_{s=1}^{|T|} \text{TF-IDF}(t_s, d_j)^2}}$$

which enforces the normalization assumption.

Keyword-based Vector Space Model

- As stated earlier, a similarity measure is required to determine the closeness between two documents. Many similarity measures have been derived to describe the proximity of two vectors; among those measures, cosine similarity is the most widely used.

$$\text{sim}(d_i, d_j) = \frac{\sum_k w_{ki} \cdot w_{kj}}{\sqrt{\sum_k w_{ki}^2} \cdot \sqrt{\sum_k w_{kj}^2}}$$

In content-based recommender systems relying on VSM, both user profiles and items are represented as weighted term vectors. Predictions of a user's interest in a particular item can be derived by computing the cosine similarity

Review of Keyword-based Systems

- Several keyword-based recommender systems have been developed in a relatively short time, and it is possible to find them in various fields of applications, **such as news, music, e-commerce, movies, etc.**
- **In the area of Web recommenders:** famous systems in literature are **Letizia , Personal WebWatcher , Syskill & Webert , ifWeb , Amalthea , and WebMate .**

Letizia is implemented as a web-browser extension that tracks the user's browsing behavior and builds a personalized model consisting of keywords related to the user's interests.

It relies on implicit feedback to infer the user's preferences. For example, bookmarking a page is interpreted as strong evidence for the user's interests in that page.

Review of Keyword-based Systems

- **Personal WebWatcher** learns individual interests of users from the Web pages they visit, and from documents lying one link away from the visited pages.
- It processes visited documents as positive examples of the user's interests, and non-visited documents as negative examples.
- **Amalthaea** uses specific filtering agents to assist users in finding interesting information as well.
- User can specify filtering agents by providing pages (represented as weighted vectors) closely related to their interests

Review of Keyword-based Systems

- In the field of **news filtering**, noteworthy recommender systems are NewT , PSUN , INFOrmer , NewsDude , Daily Learner , and YourNews.
- **NewT (News Tailor)** allows users to provide positive and negative feedback on articles, part of articles, authors or sources.
- Several filtering agents are trained for different types of information, e.g. one for political news, one for sports, etc.

Review of Keyword-based Systems

- Learning short-term and long-term profiles is quite typical of news filtering systems.
- NewsDude learns a short-term user model based on TF-IDF (cosine similarity), and a long-term model based on a naive Bayesian classifier by relying on an initial training set of interesting news articles provided by the user.
- Explore LIBRA and INTIMATE RS.

Semantic Analysis by using Ontologies

- Semantic analysis allows learning more accurate profiles that contain references to concepts defined in external knowledge bases.
- The main motivation for this approach is the challenge of providing a recommender system with the cultural and linguistic background knowledge which characterizes the ability of interpreting natural language documents and reasoning on their content.
- The description of these strategies is carried out by taking into account several criteria:
 - **the type of knowledge source involved (e.g. lexicon, ontology, etc.);**
 - **the techniques adopted for the annotation or representation of the items;**
 - **the type of content included in the user profile;**
 - **the item-profile matching strategy.**

Semantic Analysis by using Ontologies

- **SiteIF** is a personal agent for a multilingual news Web site. It was the first system to adopt a sense-based document representation in order to build a model of the user interests.
- **ITR (Item Recommender)** is a system capable of providing recommendations for items in several domains (e.g., movies, music, books), provided that descriptions of items are available as text documents (e.g. plot summaries, reviews, short abstracts).
- **SEWeP (Semantic Enhancement for Web Personalization)** is a Web personalization system that makes use of both the usage logs and the semantics of a Web site's content in order to personalize it.

